

Record Twin Field Guide

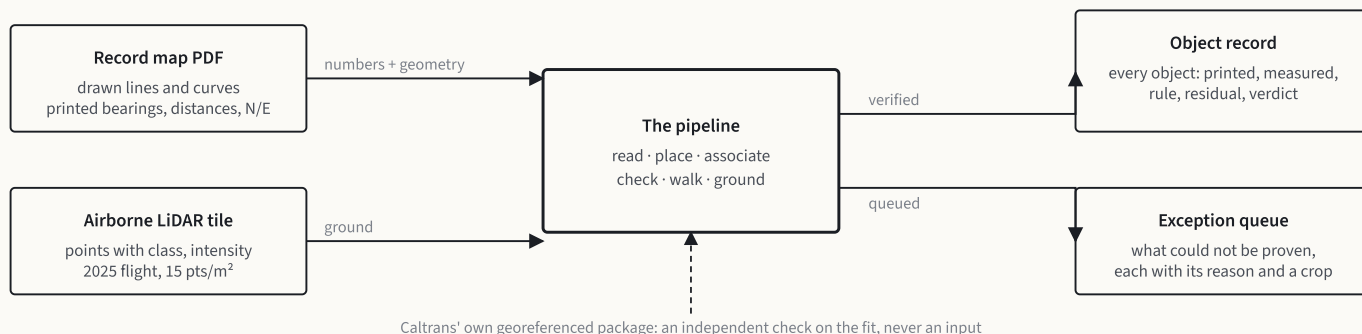
A right-of-way record map goes in. A checked, georeferenced object record comes out, on the LiDAR, with a queue of everything the system could not prove. This page teaches the machine: each section is one load-bearing mechanism, drawn, with what breaks when it is wrong.

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SECTION 1

The problem, in one picture

Caltrans District 4 holds thousands of record maps: legal drawings of what the state owns along a highway. Each sheet carries the record twice. Once as **drawn geometry** (lines, curves, circles at vertices), and once as **printed numbers** (bearings, distances, arc lengths, coordinates, areas). Nobody has ever checked one against the other at scale, and nobody has put either on the ground.



Two in, two out. The thesis is that the printed record and the drawn record must agree with each other, and both must land on the ground. Where they do, the object is verified. Where they

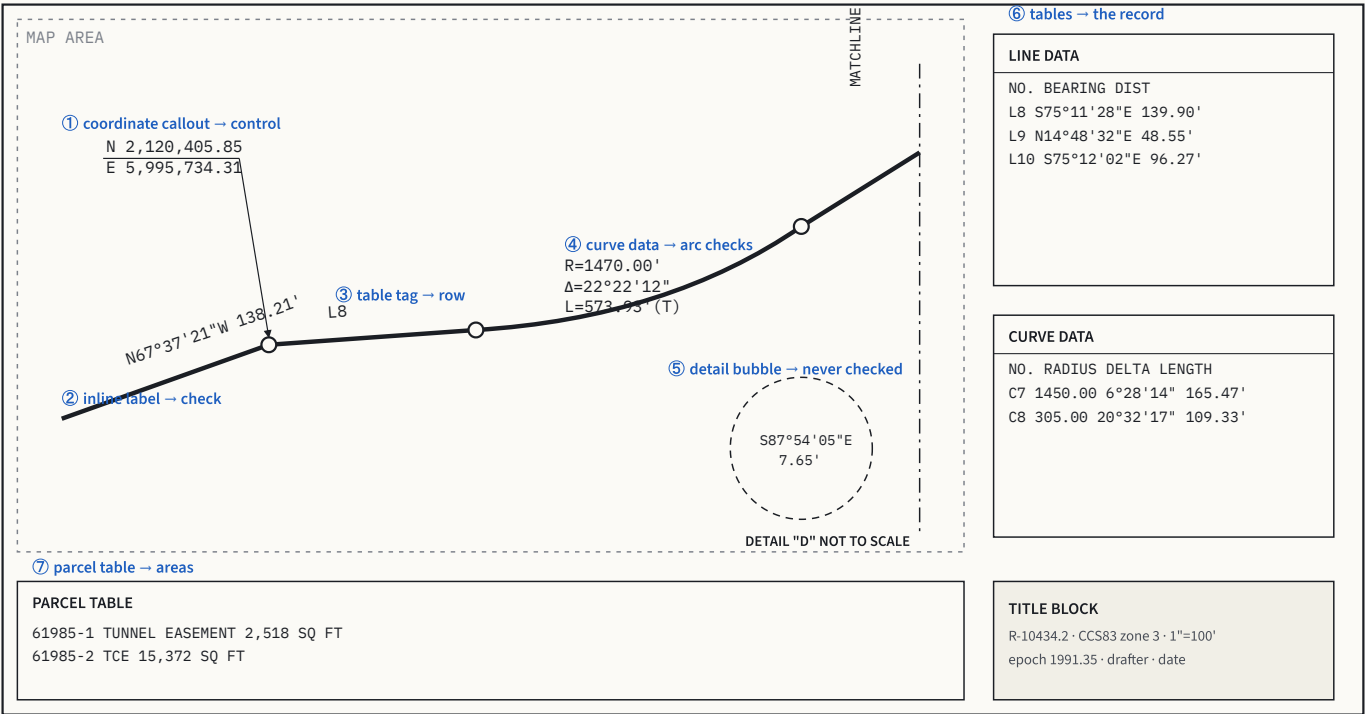
do not, it goes to a queue a surveyor can work, with the crop and the reason. Nothing is guessed.

Load-bearing: the queue is a product, not a failure log. A pass that the drawing cannot support is worse than a queued item, because a wrong "verified" stamp on a right-of-way line is the one outcome the project must never produce.

SECTION 2

Anatomy of a sheet: where the numbers live

Every mechanism downstream depends on knowing which kind of number it is looking at. A distance in the line table is not a label on the drawing; a distance inside a not-to-scale detail bubble can never match anything; a coordinate callout is control, not a check.



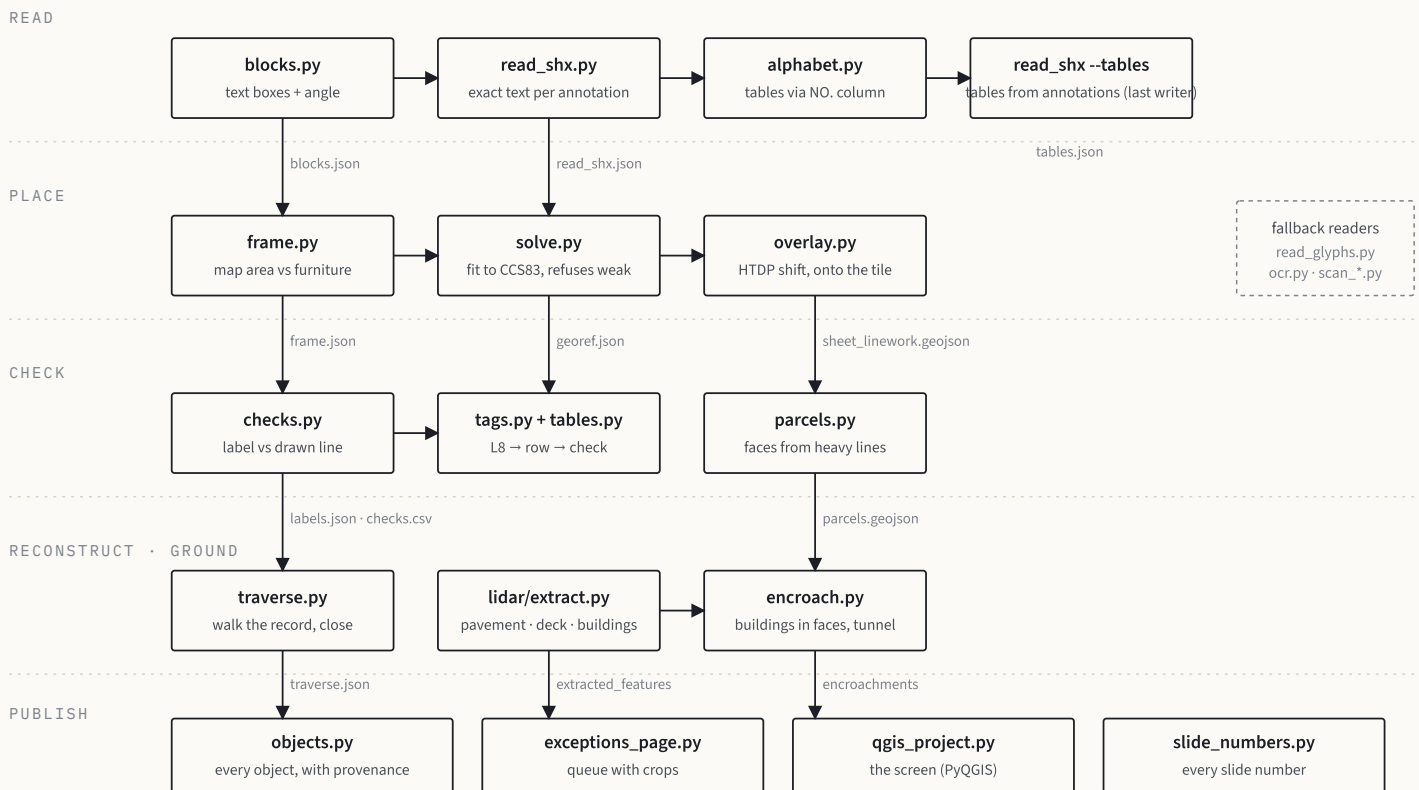
Seven kinds of number, seven fates. ① Coordinate callouts (N and E with a leader to a point) are the control that places the sheet. ② Inline labels beside a line are checked against that line. ③ Tags like L8 point to a table row, which holds the record. ④ Curve data blocks (R, Δ, L) check an arc, and check themselves ($L = R \cdot \Delta$). ⑤ Anything inside a not-to-scale bubble is queued, never failed. ⑥ The line and curve tables are the record proper; the NO. column is a known sequence, so a table validates itself. ⑦ The parcel table gives areas to compare with closed figures.

Load-bearing: masks. The map area is separated from furniture (tables, title block, legend) by the sheet's own geometry, and table cells are masked before any label check runs. Loop 4 found twelve table cells on R-10434.3 being checked as drawing labels because a mask stopped short; that alone was a third of that sheet's wrong-line count.

SECTION 3

The pipeline: what writes what

One command, offline, twenty-one scripts in a fixed order. Each script reads the files the earlier ones wrote under `spike/out/` and writes its own. The order is the dependency graph: nothing downstream re-reads the PDF's text, and nothing upstream knows about the LiDAR.



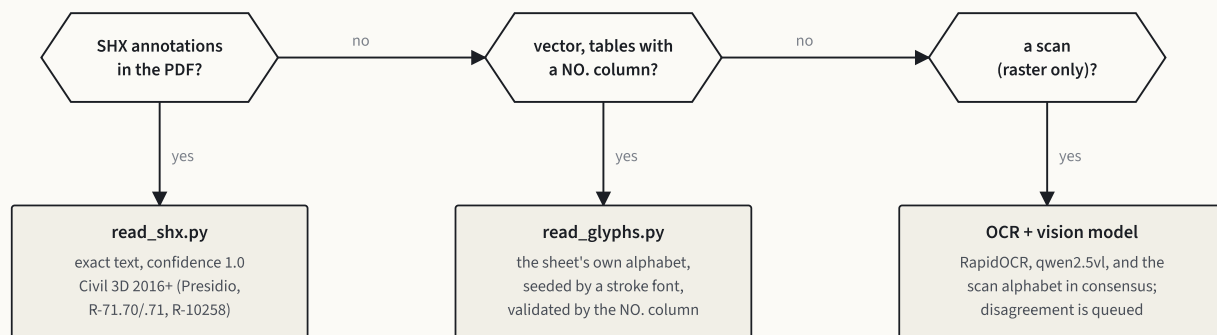
Read once, then compute. Text is read exactly once, then every check, closure and area is computed from geometry and the read files. `georef.json` is the hinge: it holds the fit every check lives in. `tables.json` has one designated last writer (annotations), because loop 3 found the demo and the bench disagreeing when two scripts wrote it in different orders.

Load-bearing: a file with two writers is a moving baseline. The two state bugs found in loop 3 (who writes `tables.json` last; a reads selector that took a 22-byte file as a validated alphabet) had shifted whole sheets' counts by tens without any logic change. Every bench since runs the canonical order.

SECTION 4

Reading text: three readers, one rule

Sheets come in three classes, and each gets the reader that is exact for it. The selector is a rule, not a score: a sheet reads by annotation where it has annotations, by its own glyph alphabet where its tables can validate that alphabet, and by OCR otherwise.



2012-era vector sheets with no tables (the north set) fall to OCR: no table can validate their alphabet

Exact where exact exists. Civil 3D 2016+ exports carry every stroked label's text as an annotation, so on those sheets reading is solved outright (Presidio: 878 annotations, 111 of 111 keyed table cells). Everything else is a fallback with a measured error rate.

How the glyph alphabet works

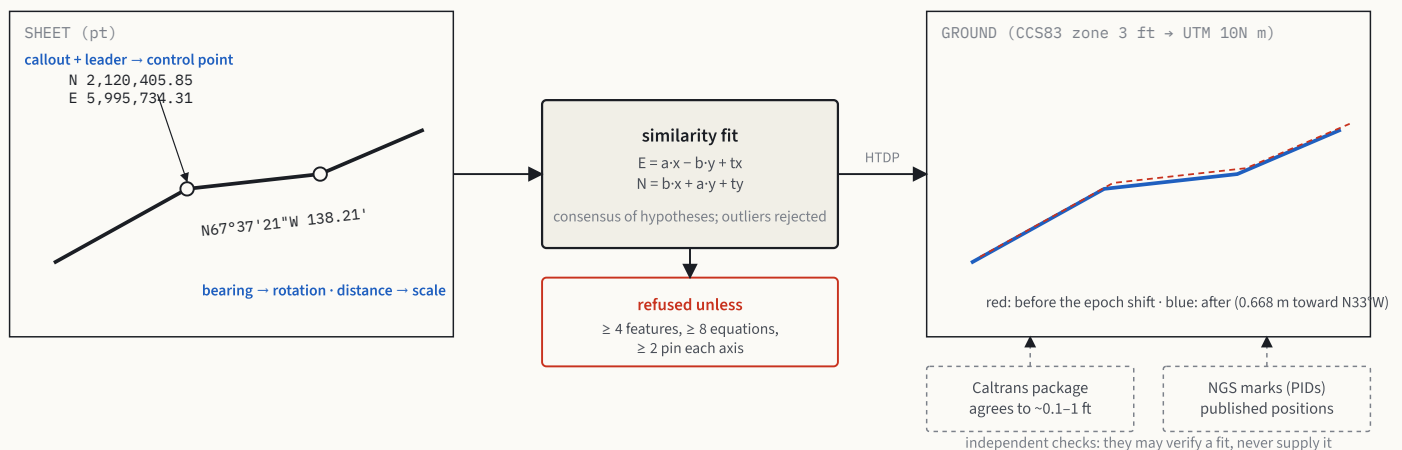
Stroked text has no characters, only paths. The trick is that a sheet's tables have a NO. column with a known sequence (L1, L2, L3...; C1, C2...). A public stroke font finds those cells, the consecutive run proves they are what they look like, and their glyphs become exact exemplars of this drafter's hand. Every other glyph on the sheet is then matched by 24 px bitmap against those exemplars at 36 rotations. A glyph matches or it is ? ; nothing reads by confidence.

Load-bearing: the reads decide the denominators of every number downstream. A label that does not read is not a fail, it is invisible; that is why the guide reports pass rates over three denominators (section 7), and why the north sheets read by OCR score high on what they check and low on what they see.

SECTION 5

Placing the sheet on the earth

Placing means one similarity transform: sheet points to ground coordinates, with scale, rotation and offset. Four things can fix it, in order of how much they prove.



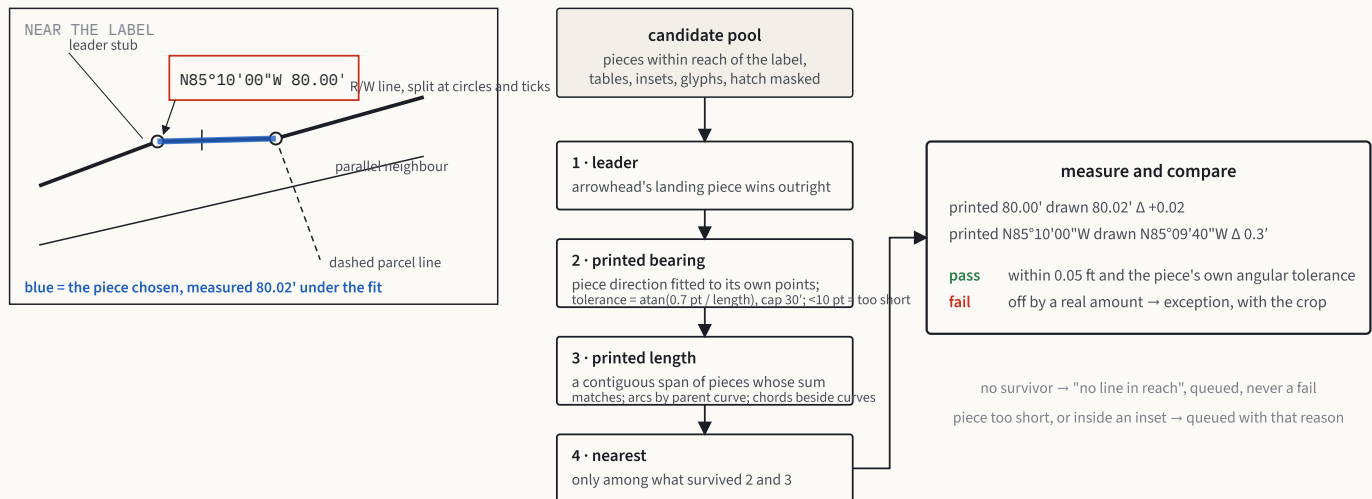
Control from the sheet's own printing. Presidio places on 18 of 19 callouts with a residual of 0.04 ft; the package agrees to about a foot, which is the package's raster limit, not ours. A sheet with no callouts still gets a *record frame* (rotation from its bearings, scale from its distances) and can be placed by the package as an offset, honestly labelled as such. A two-callout fit that the bearings confirm to 0.1° is accepted as weak, and said so.

Load-bearing: refusal. A four-point fit with 0.3 ft residuals can still be wrong by 13 % in scale; the solver refuses rather than reports, and every fit prints its disagreement with the package. Two scan fits that looked fine were caught by the grid-label scale check.

Tiles. Three LiDAR tiles are on disk (Presidio 2025, Marin 2025, a 3DEP 2018 tile for the gap between them). `tiles.py` maps a sheet to its tile by name, and the ground steps key everything off that. The epoch shift is one number for the site, from a real NGS HTDP call, cached so the demo never needs the network.

Association: which line does this number describe?

This is the wall. Reading is exact on the good sheets and placing is solved; what is hard is deciding, for a printed 138.21', which of the fifty drawn pieces near it is the one it describes. Every wrong-line exception in the project is this decision made badly. The mechanism is a candidate pool, then filters that each remove a class of wrong answers.



Filters, in the order they earn trust. A leader is the drafter telling you; the printed bearing is the record telling you; the printed length picks a span; proximity is the last resort. Each filter was added because a batch of twelve crops showed one cause dominating the wrong-line queue. Presidio's wrong-line count went 103 → 31 → 15 → 12 through that loop.

The diagnostic loop that built it

Every association rule in the codebase came from the same procedure: render the twelve largest failures with the label in red and the measured line in blue, name the cause of each, count the causes, fix the biggest, re-bench, repeat. It is cheap, it is honest, and it stops the temptation to tune thresholds by feel. Two of loop 6's findings came from crops alone: the "chord citation" was a dashed line whose dashes were 8.6 pt against an 8 pt cap, and cut pieces were inheriting their chain's seed direction.

Load-bearing: the tolerance is the drawing's, not ours. A piece 20 pt long cannot resolve a bearing better than about 2°, so the bearing tolerance scales with piece length and is capped, and pieces under 10 pt are queued as too short. Without the cap, a formula that

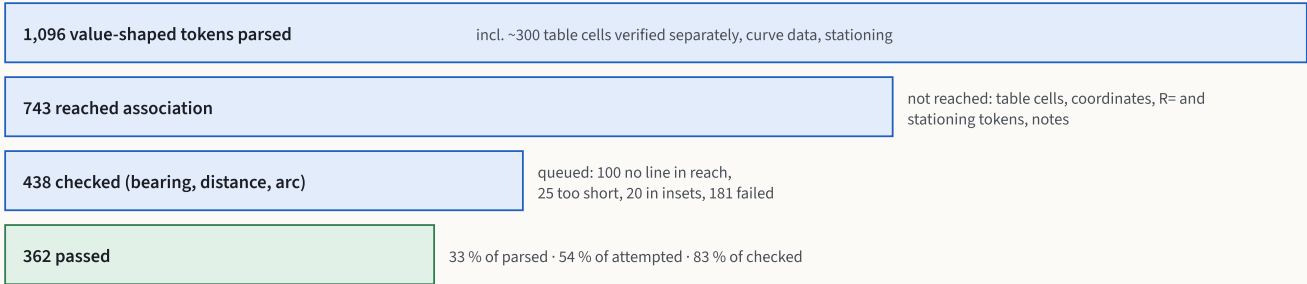
looked principled manufactured thirty passes in one leg; the cap was set from the crop captions, not from the pass count.

SECTION 7

Verdicts, the queue, and three honest denominators

Every checked label ends in one of four states, and the state names the reason. The exception page groups the non-passes by cause with a crop, a citation in PDF points, and a resolution field a surveyor fills in.

STATE	MEANING	WHERE IT GOES
PASS	printed agrees with drawn under the fit, within the drawing's own tolerance; or "pass as a run of N" where several tags share one drawn run	object record, verified
FAIL	a real disagreement between what the sheet says and what it draws	exception queue, with printed vs drawn and the difference
QUEUED	could not be checked: no line in reach, piece too short, inside a not-to-scale inset, two candidates, unread glyphs	exception queue, with the reason; never counted as wrong-line
INFO	a record with no check that applies (a coordinate, a note)	object record, as record



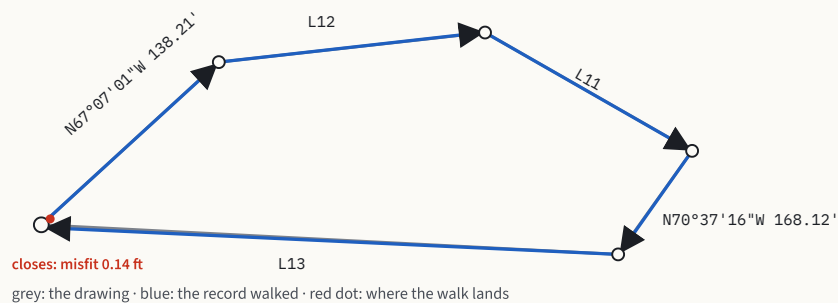
Three rates, one picture. The number to quote depends on the question. "How good is the checker?" is 83 %. "How much of the drawing's record did we verify?" is 54 %. The 33 % is over every token that looks like a value, most of which are not line labels at all. The right answer is the middle one, and the biggest single bucket behind it is "no line in reach" on the dense south sheets.

Load-bearing: a pass set, not a pass count. Every leg since loop 4 is gated on a per-label diff of which labels pass, so a change that adds three passes and silently drops three others cannot hide inside an unchanged total. Denominators are allowed to shrink only when a class of label is shown to be uncheckable, and the reason is recorded.

SECTION 8

Walking the record: the traverse

The checks say "this label agrees with this line". The traverse asks the deeper question: do the printed numbers, walked in order, come back to where they started? That is what a surveyor's closure is, and it needs no label-to-line heuristic at all once the edges are placed.



Presidio, one closed chain

edges 5 (5 with full record)
end misfit 0.14 ft
record area 6,970.6 sq ft
drawn face 6,974 sq ft (+0.06 %)
second loop: 4 edges, 0.18 ft

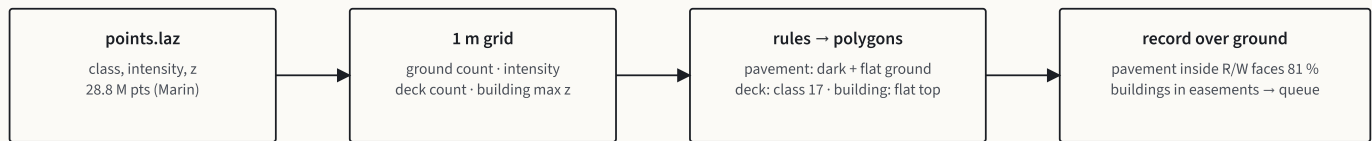
The demo's thesis in one figure. Two chains close on Presidio; none yet on the other five. Closure follows from association: every label that reaches a check becomes an edge. R-10434.1's "closed chain" from loop 2 turned out to be a transient state that never really closed, which is why the guard now drops any arc whose printed length is shorter than its own drawn chord.

The tunnel easements, settled. Parcels 61985-1..4 were expected to close as four figures. The sheet draws them as one dashed strip (345 strokes chained into eight trains), with no stroke dividing it and the curve data sitting on the R/W line. Envelope 48,410 sq ft against a table sum of 70,690: the temporary easements overlap the tunnel ones. Their outlines live in the deeds. The system measures what the sheet draws and says what it cannot prove.

SECTION 9

Ground truth from the LiDAR

Once a sheet is placed and shifted to the LiDAR epoch, its faces and lines sit on real returns. Airborne density supports three things and no more: pavement (dark, flat ground returns), viaduct deck (the bridge-deck class), and building footprints (building class with a flat top). Every feature is a raster decision on a 1 m grid then polygonised, so every polygon can be traced to the rule and the returns that made it.



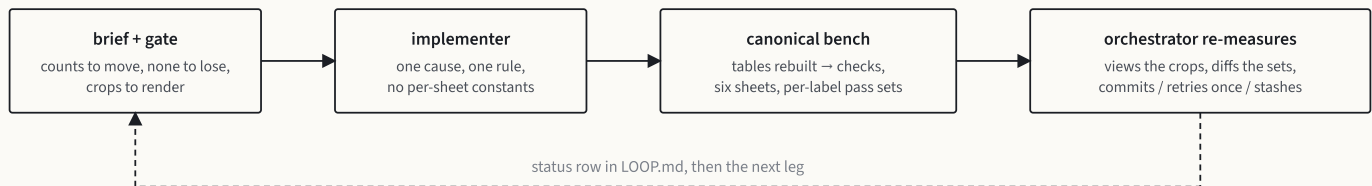
cached per tile once (grids, intensity ortho, class points); the demo never touches the LAZ again

Deterministic, traceable, bounded. No model. Curbs, signs and monuments are mobile-LiDAR work and are not claimed. Every published feature carries its rule.

SECTION 10

Measuring honestly: the bench and the gates

The project's second product is its discipline. Every change is a "leg" with a gate written before the work, measured by an orchestrator who did not write the code, on a bench whose procedure is fixed. A leg that fails its gate twice is stashed, not forced. The status table in `spike/LOOP.md` records every leg, its numbers, and what was accepted on override and why.



What counts as a pass of the gate

- the named counts moved by at least the gate, on the canonical bench, and no label lost its pass anywhere
- the implementer's crops show the blue line is the one the label describes (the orchestrator looks)
- an override is allowed only to remove fake passes or fake fails, and is written down as an override
- a hypothesis that dies is a row too (the seed font; the R-10434.1 closure): the finding is the product

Why the numbers can be trusted. Three silent regressions were caught by this loop, not by tests: two files with two writers, and a sliver guard added for one sheet that rerouted 462 annotations on another. Each was visible only because the close-out bench is canonical and diffs pass sets, not totals.

Load-bearing: the baseline row. `spike/out/bench.csv` keeps one row per sheet per leg; the current baseline is `loop6-final`. Any number quoted anywhere in the project is re-measured from a row at report time or marked unverified.

SECTION 11

The loops: what changed the numbers

Seven gated loops in three days. Each row names the mechanism that moved, not the activity.

LOOP	MECHANISM	WHAT MOVED
1 · 09-23 night	SHX annotations as the reader; association filters (bearing, span, leader) as defaults; 1969 scans on the tile via packages	Presidio distances 17/39 → 39/47; wrong-line 37 → 16
2 · 09-23 pm	Long R/W curves split at circles, lines and ticks; dash trains on the easement layer; deterministic tie-breaks	arcs 4/11 → 13/19; easements settled as one strip
3 · 09-23 eve	Six sheets on two tiles; north placed by callouts (leader trace crosses thin weights); two state bugs fixed; baseline made reproducible	the six row, twice identical
4 · 09-23 night	3DEP gap tile; two more SHX sheets placed; NGS PID control; north dashes seen (8 pt cap); table masks widened to their rows	R-10434.3 wrong-line 45 → 33; ground under all six
5 · 09-24 eve	Cut pieces take their own fitted direction; arc fallback gated to labels on curves; fillets by radius	south bearings +11; 7 fake arc fails gone
6 · 09-24 night	Lone L= sums its parent curve; bearing tolerance by piece length (capped); chords beside curves; parallels by printed bearing; insets by dashed bubble	arcs +6; bearings +16; 9 luck passes retired; two hypotheses falsified

7 · 09-25 am	The scan by its own alphabet: clean clusters, thin naming; a third reader	R-65.2 union 18/32 at 0 wrong; detection, not recognition, is the scan limit
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Where the six stand (loop6-final)

SHEET	READS	DISTANCE	BEARING	ARC	WRONG-LINE	PLACED BY
R-10434.2 Presidio	annotations	41/49	41/43	16/19	12	18 callouts, rms 0.04 ft
R-10434.1	annotations	26/37	21/23	12/24	21	13 callouts, rms 0.02 ft
R-10434.3	annotations	42/59	41/42	26/39	27	callouts, rms 0.02 ft
R-10741.1	OCR	14/18	13/13	1/2	3	3 callouts, rms 0.04 ft
R-10741.2	OCR	20/20	18/18	0/1	0	4 callouts, rms 0.05 ft
R-10741.3	OCR	17/17	13/13	0/1	1	3 callouts, rms 0.03 ft

The north reads high because OCR sees fewer labels; the south reads lower because annotations see every label, including the ones with no drawn line in reach. Same checker, different denominators: section 7 again.

SECTION 12

Glossary

record map	A right-of-way sheet: the legal drawing of what the state owns, with the survey record printed on it.
bearing · distance	A line's direction (N67°37'21"W: 67°37'21" west of north) and length in US survey feet. Printed beside the line or in a table row.
curve data	R (radius), Δ (central angle), L (arc length) for a curve. L = R·Δ is an internal check that needs no drawing.
(T)	A total: the printed length of a whole run that several tags share. Checked as a run-sum.
callout	A printed N/E coordinate with a leader to the point it locates. The sheet's own control.
leader · arrowhead	The thin line from a label to the thing it describes, ending at a filled triangle or a vertex circle.

SHX annotation	AutoCAD's text as an annotation object in the PDF. Exact text, exact rectangle. Present on Civil 3D 2016+ exports.
glyph alphabet	Per-sheet exemplars of the drafter's stroked characters, seeded by a public font and validated by the tables' NO. column.
CCS83 zone 3	California Coordinate System, NAD83, zone 3, US survey feet: what the sheets print. Converted to UTM 10N metres for the LiDAR.
HTDP	NGS's Horizontal Time-Dependent Positioning: the epoch shift between the sheet's datum epoch and the LiDAR's. 0.668 m toward N33°W here.
package	Caltrans' own georeferenced raster of a sheet (tif + tfw) from the D4 index. Used as a check on the fit, never as an input.
wrong-line	An exception whose disagreement is far too large to be a drafting error: the reader measured a different line. The main diagnostic.
canonical bench	The fixed procedure that produces a bench row: annotation tables rebuilt, then checks, on all six sheets, per-label pass sets diffed.
leg · gate · loop	One change, its acceptance criterion written first, and the sequence of legs run overnight or in an evening with an orchestrator re-measuring each.
tile	One LiDAR delivery: points.laz + DEM. Three on disk; a sheet is routed to its tile by name.

SOURCES: STATE.MD, ROADMAP.MD, SPIKE/LOOP.MD,
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